BLUE RIDGE HEALTH DISTRICT, VIRGINIA · PROJECT PACKET

Blue Ridge Maternal Access Map

Which communities are both farthest from labor and delivery care and most economically disadvantaged? This packet covers the findings, heat maps, access profiles, CHIP overlay, telehealth and AI implications, the six locality one-pagers, resume bullets and interview notes.

Prepared by Neha Vittal · October 2026 · Independent project

INTERACTIVE MAP (HTML)

https://claude.ai/artifact/75HKTjc2Pt58CZZTzk9wwD

Seven switchable heat-map layers, a sortable tract heat grid, access profiles, the CHIP overlay, the telehealth chart and the one-pagers. The link is private until it is shared from the page's Share menu.

1 in 3 district residents (about 87,300) live 30+ minutes from the nearest labor and delivery unit	**66,400** residents in 15 tracts that are both far from delivery care and more disadvantaged	**0 of 4** rural counties (Fluvanna, Greene, Louisa, Nelson) have a hospital that delivers babies	**42,600** residents live where the drive is long and 15%+ of homes have no internet

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Distance and disadvantage stack up in the rural counties

> "I asked which Blue Ridge communities are both farthest from labor and delivery care and most disadvantaged. I combined deprivation, drive time, WIC and internet data for 64 census tracts, grouped them into four access profiles, and checked the health district's priority tracts against the result."

Scope

64 census tracts (2020 boundaries), about 264,000 residents, in Albemarle, Charlottesville, Fluvanna, Greene, Louisa and Nelson.

Key findings

- **About 66,400 residents live in 15 "remote and disadvantaged" tracts.** These pair long drives to delivery care (median 47 min) with above-median deprivation (median ADI 50). All 15 are in Fluvanna, Greene, Louisa and Nelson. None of those counties has a hospital that delivers babies.
- **Distance:** 1 in 3 residents live 30+ minutes from the nearest labor and delivery unit, and 1 in 6 live 45+ minutes away. The share over 45 minutes is 73% in Nelson, 50% in Louisa, 18% in Fluvanna and 12% in Greene.
- **Most burdened tract:** Arrington-Wingina in Nelson has the district's highest deprivation score (78th national percentile) and averages a 61-minute drive.
- **Regional context:** Centra Southside Community Hospital in Farmville stopped delivering babies on December 19, 2025.

What it means for the CHIP

- **The priorities mostly hold up.** Four of the six CHIP tracts (Arrington-Wingina, Town of Louisa, Stanardsville, Columbia/Fork Union) are confirmed once distance is added. Three of them are the top three tracts on double burden.
- **The two urban CHIP tracts need a different fix.** Branchlands/Squire Hill and Fifeville are 2 to 10 minutes from a hospital, but 22% and 16% of households have no car.
- **Deprivation alone misses places.** Six tracts rank in the top 10 on double burden but were not prioritized: Trevillians, Roseland, Mineral, Lovingston, Monroe and Gum Springs–Holly Grove.

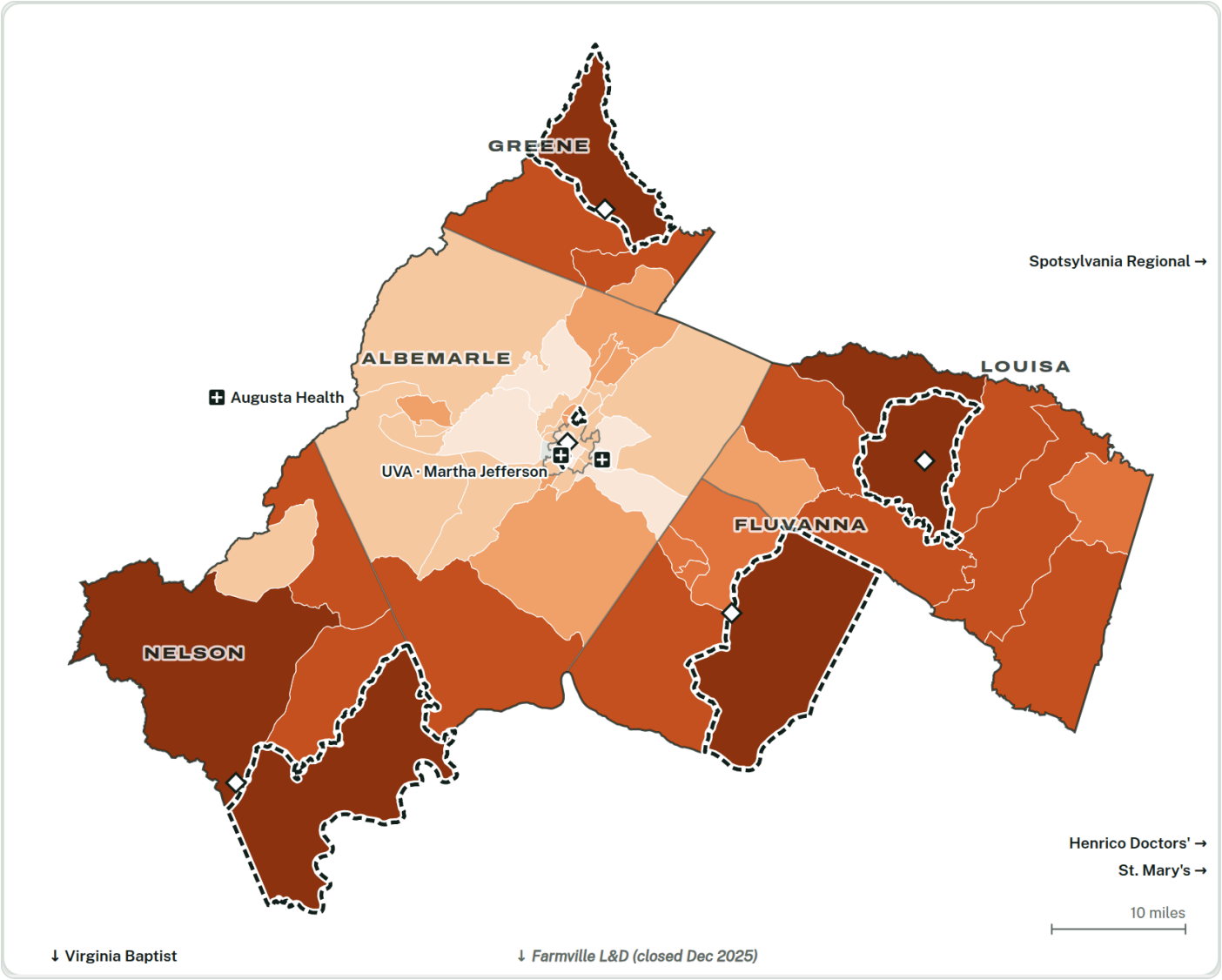
Telehealth and AI in one line each

- **Telehealth** only helps where homes can connect. About 42,600 residents live where the drive is long and 15%+ of homes are offline, so the plan there has to start with phone visits and cellular blood-pressure cuffs.
- **AI** targets and extends care. It can't open a birthing unit or shorten a 61-minute drive during labor.

2 · HEAT MAP

Double burden: far from delivery care and more disadvantaged

A tract ranks high only when it is high on both deprivation and drive time. The score is the geometric mean of each tract's percentile rank on ADI and on drive time to labor and delivery (0–100).

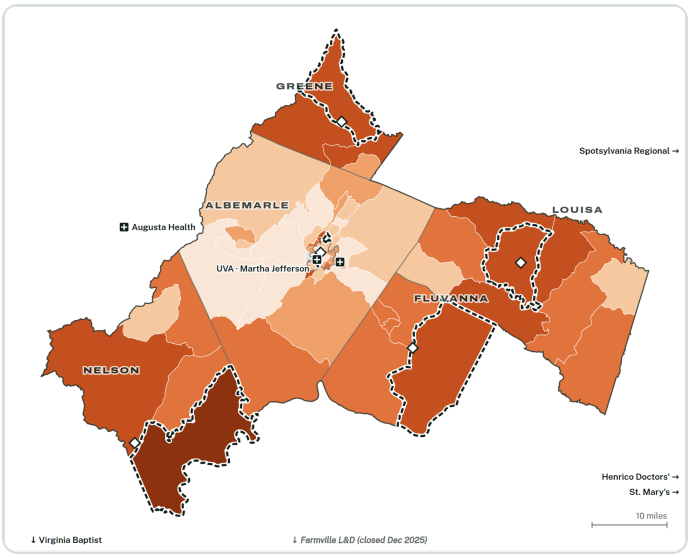


■ with cross: labor and delivery unit ◆ WIC site Dashed outline: CHIP priority tract

Arrows: hospitals beyond the map edge

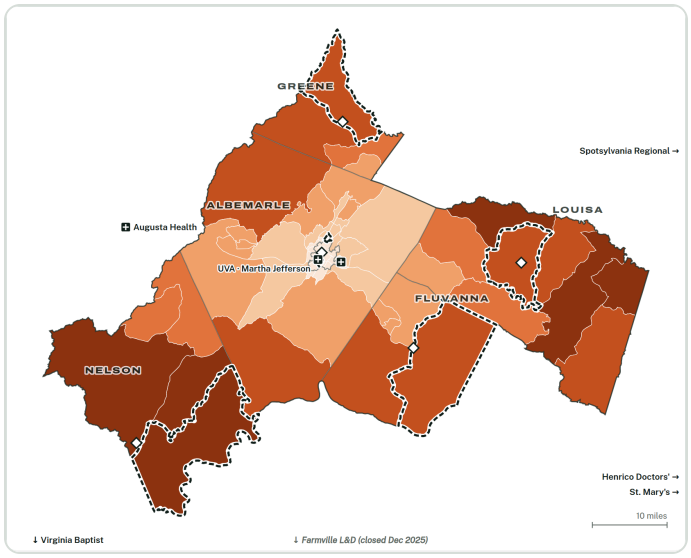
Top five tracts: Arrington-Wingina (Nelson, CHIP), Town of Louisa (CHIP), Stanardsville (Greene, CHIP), Trevillians (Louisa), Columbia/Fork Union (Fluvanna, CHIP).

The four measures behind the score



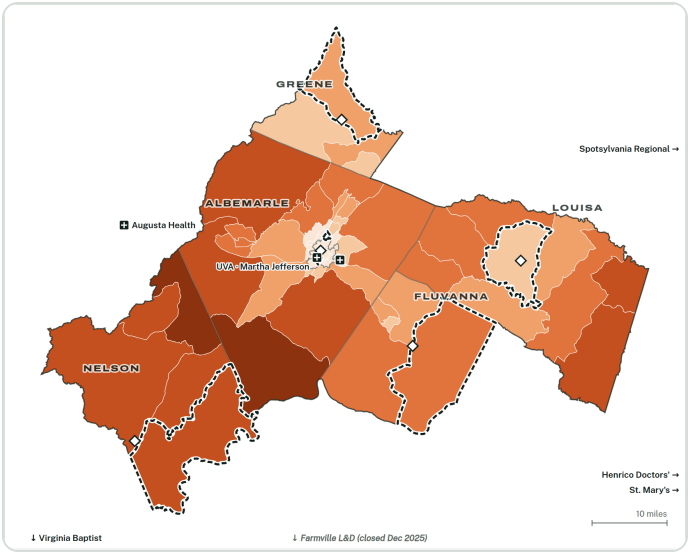
Deprivation (ADI, 2021). National percentile; 100 = most disadvantaged.

< 20 | 20–30 | 30–40 | 40–50 | 50–60 | 60+



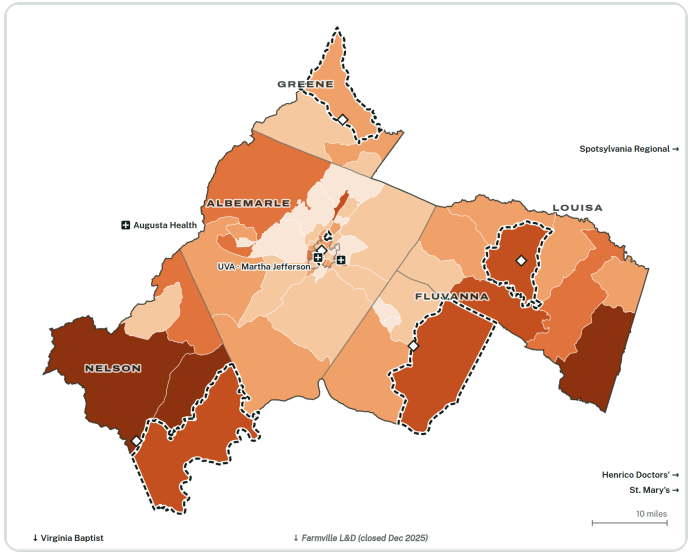
Drive to labor and delivery (minutes). Free-flow, from where people live.

< 10 | 10–20 | 20–30 | 30–40 | 40–50 | 50+



Drive to WIC (minutes). Closest of the five BRHD health department offices.

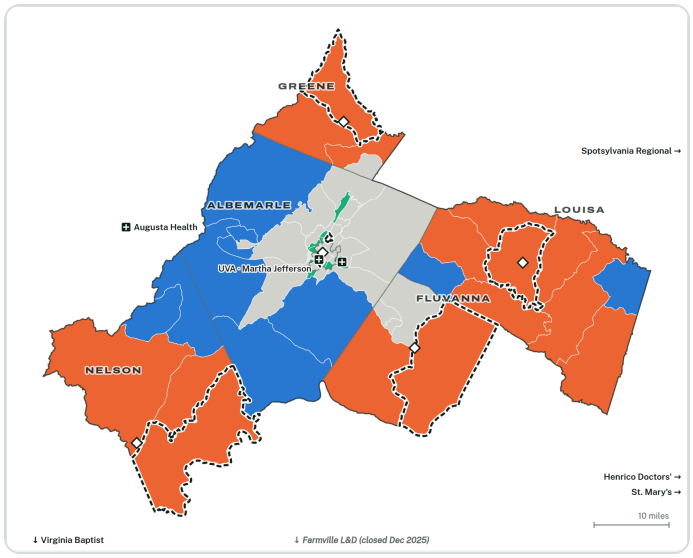
< 10 | 10–15 | 15–20 | 20–30 | 30–40 | 40+



Households with no internet (%). Stands in for broadband access (CDC SVI 2022).

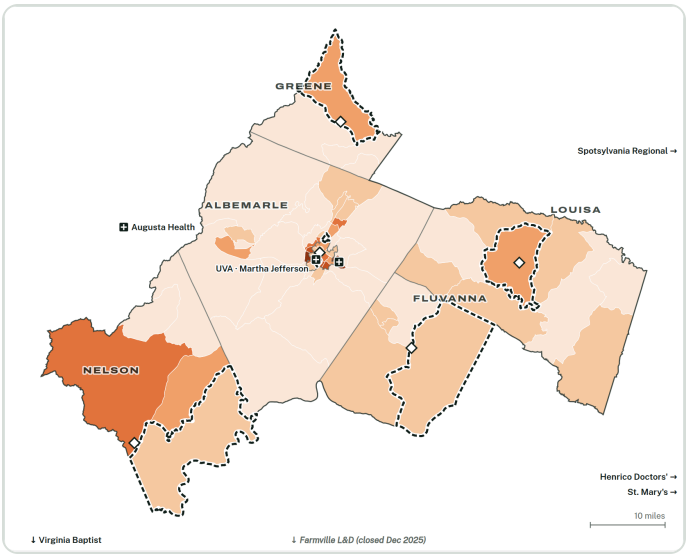
< 5 | 5–10 | 10–15 | 15–20 | 20–25 | 25+

Four access profiles



Access profiles. K-means on ADI, drive to L&D, drive to WIC and internet access.

- A · Remote and disadvantaged
- B · Rural, long drives, fewer economic barriers
- C · Near care, higher deprivation
- D · Near care, lower deprivation



Households with no vehicle (%). Shown as its own layer; it explains the urban CHIP tracts.

< 2 | 2–5 | 5–10 | 10–15 | 15–20 | 20+

Profile	Tracts	Residents	Median ADI	L&D drive	WIC drive	No internet
A · Remote and disadvantaged	15	66,436	50	47 min	22 min	17%
B · Rural, long drives, fewer economic barriers	10	38,780	27	34 min	35 min	13%
C · Near care, higher deprivation	14	47,883	45	6 min	7 min	15%
D · Near care, lower deprivation	24	104,316	23	15 min	14 min	5%

Why four groups: k = 4 was the most stable under bootstrap resampling (mean ARI 0.82 over 200 resamples) and nearly matched Ward hierarchical clustering (ARI 0.97), with silhouette 0.35 (best was 0.38 at k = 2). Adding households without a vehicle as a fifth input kept most assignments (ARI 0.83). One tract (Carr's Hill–McCormick Road, UVA student housing) has no ADI and is not clustered.

5 · HEAT GRID AND CHIP OVERLAY

Every tract, every measure

Tract	Profile	Double burden	ADI (nat. pct.)	Drive to L&D (min)	Drive to WIC (min)	No internet %	No vehicle %	Poverty %	Residents
Wintergreen-Wingina Nelson · CHIP priority	A	100	78	61	32	23.4	4.3	14.7	3,185
Town of Louisa Louisa · CHIP priority	A	92	59	47	14	23.1	9.7	10.2	5,659
Stanardsville Greene · CHIP priority	A	92	56	49	16	13.9	5.2	16.7	4,035
Trevillians Louisa	A	90	52	53	22	14.8	2.2	3.2	3,291
Columbia/Fork Union Fluvanna · CHIP priority	A	89	54	47	23	22.5	2.5	12.7	5,167
Roseland Nelson	A	86	50	52	30	29.3	14.2	16.6	3,838
Mineral Louisa	A	84	51	47	17	14.7	2.7	8.0	4,771
Lovingston Nelson	A	82	47	56	35	25.1	5.3	7.8	2,522
Monroe Greene	A	80	50	45	12	6.3	0.0	18.5	4,628
Gum Springs-Holly Grove Louisa	A	78	49	45	34	27.3	4.7	14.6	6,920
Ferncliff-Shannon Hill (Yanceyville) Louisa	A	78	53	38	17	23.8	4.7	14.9	3,317
Cuckoo District Louisa	A	77	44	51	24	17.3	0.0	11.5	5,214

Top of the heat grid, sorted by double burden. Cells use the map's color classes; a dark bar marks CHIP priority tracts. The interactive version lists all 64 tracts.

How the six CHIP priority tracts compare

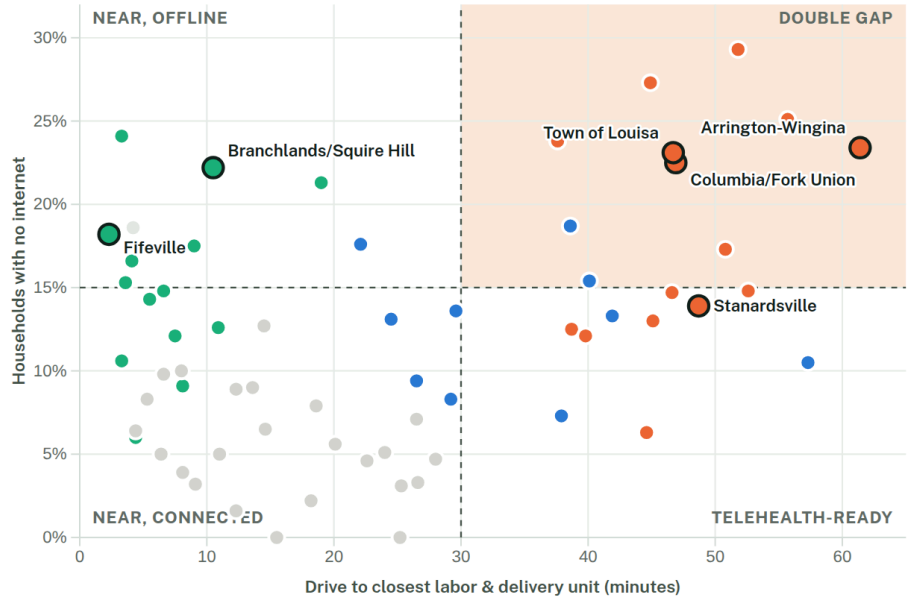
CHIP tract	Profile	Double burden	ADI (nat. pct.)	Drive to L&D (min)	Drive to WIC (min)	No internet %	No vehicle %	Burden rank
Arrington-Wingina Nelson	A	100	78	61	32	23.4	4.3	#1
Town of Louisa Louisa	A	92	59	47	14	23.1	9.7	#2
Stanardsville Greene	A	92	56	49	16	13.9	5.2	#3
Columbia/Fork Union Fluvanna	A	89	54	47	23	22.5	2.5	#5
Branchlands/Squire Hill Albemarle	C	53	54	11	8	22.2	22.1	#22
Fifeville Charlottesville	C	12	56	2	6	18.2	15.5	#60

ADI check: using the 2021 ADI, the CHIP tract is the highest-ADI tract in Fluvanna, Greene, Louisa and Nelson. In Albemarle and Charlottesville it ranks 3rd and 2nd, within 3 percentile points of the top. A newer ADI release or a different aggregation likely explains the gap.

Telehealth only helps where homes can connect

Drive to L&D vs. homes with no internet

Each dot is a census tract, colored by access profile. Dashed lines mark 30 minutes and 15% of homes. CHIP tracts are outlined and labeled.



Group (30 min · 15% offline)	Tracts	Residents	Homes offline	What telehealth means
Double gap: far and offline	10	~42,600	~3,900	Video visits miss many homes. Start with audio-only visits (covered by Virginia Medicaid since 2022), cellular blood-pressure cuffs, hotspot lending or a telehealth room at the health department. CHIP tracts: Arrington-Wingina, Town of Louisa, Columbia/Fork Union.
Telehealth-ready: far but connected	10	~41,700	~1,800	Biggest time savings. Move routine prenatal check-ins and postpartum blood-pressure checks to video and home monitoring. Tailored schedules use 8–10 visits instead of 12–14. CHIP tract: Stanardsville.
Near care, offline	9	~35,000	~2,300	The barrier is cars and connection, not distance. Phone visits save bus trips. CHIP tracts: Branchlands/Squire Hill, Fifeville.
Near care, connected	35	~145,000	~4,200	Standard hybrid prenatal care fits.

About 2 hours

Round trip from Arrington-Wingina to the closest delivering hospital. Each visit moved to telehealth saves that drive.

Still 61 minutes

The same drive during labor. Telehealth can't move birth, ultrasounds, labs or emergencies.

Cost matters

The federal Affordable Connectivity Program ended in June 2024. Virginia's 2025 broadband study names cost as the top reason households go without internet.

What AI can and can't do here

AI can't open a birthing unit or shorten the drive during labor. It can help decide where limited outreach, devices and staff go, explain findings plainly, and keep contact with families between visits.

Job	What AI does	Evidence	Limits
Find the gaps BUILT HERE	Machine-learning clustering (k-means) sorts tracts into four access profiles, so each profile gets a matching strategy.	This project: stable at k = 4 (bootstrap ARI 0.82; 0.97 agreement with Ward clustering).	Only as good as its inputs. ACS margins are wide in small tracts; nearest hospital is not always where people deliver.
Explain it plainly BUILT HERE	A large language model drafts a one-pager for each locality using only a fact sheet of computed numbers.	This project: all six drafts at grade 5.8–7.9 reading level, each checked line by line against the tract table.	An unconstrained model can invent numbers or programs. Keep the fact-sheet-only rule and human review.
Watch between visits NEXT STEP	Text-based home blood-pressure programs flag dangerous readings after delivery. Automation or AI can triage readings so nurses see urgent ones first.	Penn's Heart Safe Motherhood (randomized trial): 92% sent a reading vs. 44% who came in for an office check, and it closed the racial gap in follow-up. Virginia Medicaid's remote-monitoring benefit for high-risk pregnancy explicitly includes hypertension and diabetes since July 2025.	Most proven programs are rules-based, not AI. Offline homes need cellular cuffs. A clinician must be on call.
Bring scans closer NEXT STEP	AI-guided "blind sweep" ultrasound lets staff without sonography training estimate gestational age with a low-cost handheld probe.	2024 study in Zambia and North Carolina: after one day of training, AI error averaged 3.2 days vs. 3.0 days for expert sonographers.	Dating only, not a full anatomy scan. Not routine US practice; needs OB oversight and a referral path.
Help families navigate NEXT STEP	A text assistant could help book Medicaid rides and WIC appointments and build a labor travel plan.	Untested in this district; pilot with community health workers.	Needs fast handoff to a person, privacy protection, and plain-text access for offline homes.
Check for bias ALWAYS	Audit any risk score or triage model by race, rurality and insurance before it steers care.	The national VBAC calculator used race to predict lower success for Black and Hispanic patients until it was revised in 2021.	A model trained on who used care can miss the people who couldn't reach it.

> Bottom line: AI targets and extends care; it does not replace it. In the double-gap tracts, the plan has to work by phone and cellular device first. The drive at delivery stays until the district adds birthing capacity, transport and emergency readiness.

How it was built

1. **Tracts and census measures.** 64 census tracts (2020 boundaries) with ACS 2020–24 estimates from the UVA Equity Center's Regional Equity Dashboard data.
2. **Deprivation.** Area Deprivation Index, 2021 release (national percentile). In this release, block groups inherit the tract value.
3. **Broadband and vehicles.** Households with no internet subscription and with no vehicle, from CDC/ATSDR SVI 2022 (ACS 2018–22), with margins of error kept.
4. **Drive time.** OSRM car routing on OpenStreetMap roads, free-flow. Small tracts routed from the population-weighted centroid; tracts over 15 sq mi routed from every block group and averaged by population. Town-center checks to UVA: Lovingston 42 min (31 mi), Louisa 43 min (32 mi), Stanardsville 38 min (23 mi).
5. **Facilities.** Twelve hospitals that currently deliver babies, checked in September 2026; Farmville shown as closed. WIC sites are the five BRHD health department offices.
6. **Clustering.** K-means on z-scored inputs, k = 4.
7. **One-pagers.** Drafted with an LLM from locality fact sheets, checked against the tract table, scored for reading level.

Limits

- Measures access, not outcomes: no birth or mortality data yet.
- Nearest is not always where people deliver (insurance, provider, risk level).
- Drive times assume a car and no traffic.
- "No internet subscription" measures adoption, not availability.
- ACS tract estimates have wide margins in small tracts.
- Nelson WIC location placed from its address (within ~1 km).

Next steps

- Add counts of women aged 15–44 (Census API key).
- Link to birth outcomes and severe maternal morbidity data.
- Add FCC broadband availability as a second layer.
- Review the map with BRHD partners and community health workers.

Sources

- BRHD 2025–2028 Community Health Improvement Plan
- UVA Equity Center, Regional Equity Dashboard data
- Neighborhood Atlas, Area Deprivation Index (2021 file via geocode-adi)
- CDC/ATSDR Social Vulnerability Index 2022
- OSRM routing on OpenStreetMap data
- BRHD health department locations
- Virginia Mercury: Farmville L&D closure
- Kaiser Permanente: telehealth plus in-person prenatal care outcomes
- U-M IHPI: tailored prenatal care guidelines
- Heart Safe Motherhood evidence
- DMAS: audio-only coverage · DMAS: maternal health changes, July 2025
- Virginia 2025 broadband affordability study
- AI blind-sweep ultrasound accuracy study
- STAT: race removed from the VBAC calculator
- Virginia Navigator: Medicaid non-emergency transportation

Resume bullets

Blue Ridge Maternal Health Access Map | Independent project | Python (geopandas, scikit-learn), D3, LLMs | 2026

Full version

- Built a census-tract map of maternal care access for Virginia's Blue Ridge Health District (64 tracts, ~264,000 residents), combining the Area Deprivation Index, ACS and CDC broadband and vehicle data, WIC sites, and drive times to 12 labor and delivery units.
- Calculated population-weighted drive times from 186 block groups on OpenStreetMap roads; found 1 in 3 residents live 30+ minutes from the nearest labor and delivery unit, including 73% of Nelson County residents who live 45+ minutes away.
- Used k-means clustering to group tracts into 4 access profiles, checked with silhouette scores, bootstrap stability (ARI 0.82) and a second clustering method (0.97 agreement); identified ~66,400 residents in 15 "remote and disadvantaged" tracts.
- Tested the district's 6 CHIP priority tracts against a new distance-plus-deprivation index: confirmed 4, showed the 2 urban tracts face car-access barriers rather than distance, and flagged 6 high-burden tracts the plan did not prioritize.
- Assessed telehealth readiness and found ~42,600 residents in tracts that are both far from delivery care and 15%+ offline; recommended phone-first telehealth and cellular blood-pressure monitoring for those areas.
- Used an LLM to draft 6 plain-language one-pagers from verified fact sheets, checked every number by hand, and scored them at a grade 6–8 reading level; published results as an interactive D3 heat-map dashboard.

Short version

- Mapped maternal care access across 64 census tracts (~264,000 residents) using deprivation, broadband and WIC data and routed drive times to labor and delivery units; found 1 in 3 residents live 30+ minutes from delivery care.
- Clustered tracts into 4 access profiles with k-means (bootstrap-validated); showed ~66,400 residents face both long drives and high deprivation, and flagged 6 high-burden tracts missing from the district's CHIP priorities.
- Turned the findings into an interactive D3 dashboard and 6 plain-language one-pagers drafted with an LLM (grade 6–8 reading level), plus a telehealth readiness analysis.

Telehealth and AI bullets

- Grouped 64 census tracts by drive time to labor and delivery and home internet access to plan telehealth; found ~42,600 residents in 10 tracts where video visits would miss 15%+ of homes and recommended phone-first visits (covered by Virginia Medicaid) and cellular blood-pressure cuffs there.
- Identified ~41,700 residents in far-but-connected tracts where moving routine prenatal and postpartum blood-pressure checks to telehealth could save round trips of up to 2 hours per visit, in line with tailored prenatal care guidance (8–10 visits plus home monitoring).
- Built a guarded LLM workflow that drafted 6 plain-language community one-pagers only from verified fact sheets, with every number checked by hand and a grade 6–8 reading level.
- Wrote an evidence-based AI roadmap for maternal care access (remote blood-pressure triage, AI-guided ultrasound, text-message navigation for rides and WIC), with evidence, limits and a bias-audit step for each use.

One line for LinkedIn

Mapped where distance and disadvantage overlap for maternal care in Virginia's Blue Ridge region, using routed drive times, k-means clustering and AI-drafted one-pagers for the community.

Interview key takeaways

30-second pitch

> "I asked which Blue Ridge communities are both farthest from labor and delivery care and most disadvantaged. I combined deprivation, drive time, WIC and internet data for 64 census tracts, grouped them into four access profiles, and checked the health district's priority tracts against the result. The finding is that distance and disadvantage stack up in the rural counties, and that telehealth only helps where homes can connect."

Numbers to remember

- **Scope:** 64 tracts, ~264,000 residents, six localities.
- **Distance:** 1 in 3 residents live 30+ minutes from L&D; none of the four rural counties has a unit.
- **Most burdened:** ~66,400 residents in 15 tracts.
- **Worst tract:** Arrington-Wingina, 61-minute drive, highest deprivation.
- **Telehealth:** ~42,600 residents far from care and 15%+ offline.

Three findings

- **Priorities mostly hold up:** 4 of 6 CHIP tracts confirmed once distance is added.
- **Urban CHIP tracts need a different fix:** minutes from a hospital, but 22% and 16% of households have no car.
- **Deprivation alone misses places:** 6 high-burden tracts, including Trevillians and Roseland, were not prioritized.

Methods, in plain words

- **Drive times:** real roads, not straight lines; rural tracts averaged across block groups, since one center point can land on a back road and add 20–30 minutes.
- **Clustering:** k-means on four inputs; four groups were the most stable when re-run on resampled data, and a second method agreed.
- **Combined score:** high only if high on both deprivation and drive time.
- **AI drafting:** computed numbers only, checked by hand, grade 6–8.

Your stance on AI and telehealth

- AI targets and extends care; it can't open a birthing unit or shorten a 61-minute drive in labor.
- Telehealth depends on connection: offline areas need phone visits and cellular blood-pressure cuffs first.
- Any risk model needs a bias check by race, rurality and insurance.

Limits to name before they ask

- Access, not outcomes: no birth or mortality data yet.
- Closest hospital isn't always where people deliver.
- Drive times assume a car and no traffic.
- Next: women aged 15–44 counts, birth outcomes, partner review.

If they ask how you built it

- Independent project. You used an AI assistant for code and drafts; you set the question, chose the methods and checked the numbers.
- **Background tie:** extends your SparkMap work at MDHHS; tailored prenatal care (MiPATH) came out of Michigan.

BIRTH CARE ACCESS · NELSON COUNTY · BLUE RIDGE HEALTH DISTRICT

Nelson County: the longest drive to give birth

No hospital in Nelson County delivers babies. Most families drive 45 minutes or more to reach one. The area around Arrington and Wingina has the longest drive and the most money hardship in the whole health district.

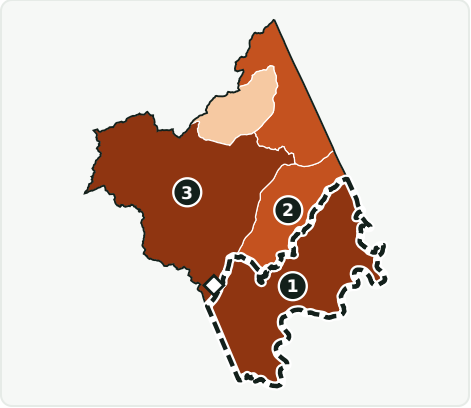
50 min
average drive to the closest hospital that delivers babies

73%
of residents live more than 45 minutes away

21%
of homes have no internet service

WHAT WE FOUND

- The closest birth hospital depends on where you live: UVA in Charlottesville, Augusta Health in Fishersville, or Virginia Baptist in Lynchburg.
- The drive to WIC at the Nelson County Health Department averages 35 minutes.
- About 1 in 5 homes has no internet. That makes video visits with a doctor or WIC hard.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

PLACES TO WATCH

1. **Arrington-Wingina**
   Drive to a birth hospital: about 61 minutes. Hardship score (ADI): 78 out of 100, the highest in the district. This is the county's CHIP priority area.
2. **Lovingston**
   Drive: about 56 minutes, with 1 in 4 homes offline.
3. **Roseland**
   Drive: about 52 minutes. 29% of homes have no internet and 14% have no car.

WHAT COULD HELP

- Make a travel plan for labor during prenatal visits: which hospital, who drives, and a backup ride.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Offer check-ins by phone, not just video, for families without internet.
- Bring WIC closer to home with scheduled pop-up days in Arrington, Lovingston, and Roseland.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 6.9 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

BIRTH CARE ACCESS · LOUISA COUNTY · BLUE RIDGE HEALTH DISTRICT

Louisa County: half of residents live 45+ minutes from birth care

No hospital in Louisa County delivers babies. Nearly everyone lives more than 30 minutes from one, and half live more than 45 minutes away. The Town of Louisa has both a long drive and the most money hardship in the county.

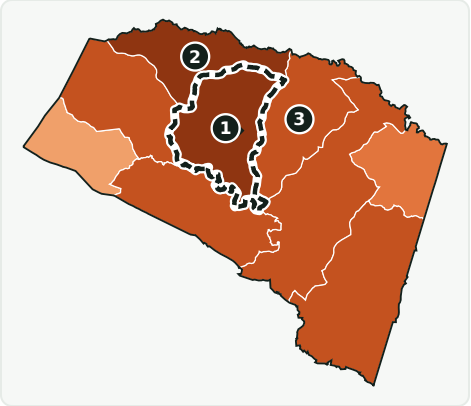
45 min
average drive to the closest hospital that delivers babies

96%
of residents live more than 30 minutes away

18%
of homes have no internet service

WHAT WE FOUND

- Families here are split between four hospitals. Martha Jefferson in Charlottesville is closest for about half. The others are Spotsylvania Regional, St. Mary's, and Henrico Doctors'.
- The drive to WIC in the Town of Louisa averages 24 minutes.
- 7 of the county's 9 neighborhoods (census tracts) fall in the "remote and disadvantaged" group.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

PLACES TO WATCH

1. **Town of Louisa**
   Drive to a birth hospital: about 47 minutes. Hardship score (ADI): 59 out of 100. 23% of homes are offline. This is the county's CHIP priority area.
2. **Trevillians**
   Drive: about 53 minutes, with a hardship score of 52. Not a CHIP priority area, but it ranks 4th in the district.
3. **Mineral**
   Drive: about 47 minutes, with a hardship score of 51.

WHAT COULD HELP

- Help each family choose a birth hospital early and plan the route, since the closest one differs across the county.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Offer check-ins by phone for homes without internet.
- Include Trevillians and Mineral in outreach, not only the Town of Louisa.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 6.3 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

BIRTH CARE ACCESS · GREENE COUNTY · BLUE RIDGE HEALTH DISTRICT

Greene County: close to Charlottesville, still a long drive

No hospital in Greene County delivers babies. Most families go to UVA in Charlottesville. For about 6 in 10 residents, that drive is more than 30 minutes. Stanardsville has the longest drive and the most money hardship.

37 min
average drive to the closest hospital that delivers babies

63%
of residents live more than 30 minutes away

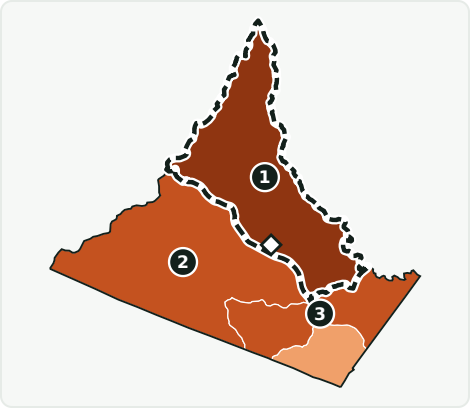
15 min
average drive to WIC in Stanardsville

WHAT WE FOUND

- UVA in Charlottesville is the closest birth hospital for every neighborhood in the county.
- WIC at the Greene County Health Department is fairly close for most families.
- 3 of the county's 4 neighborhoods fall in the "remote and disadvantaged" group.

PLACES TO WATCH

1. **Stanardsville**
   Drive to a birth hospital: about 49 minutes. Hardship score (ADI): 56 out of 100. This is the county's CHIP priority area and ranks 3rd in the district.
2. **Monroe**
   Drive: about 45 minutes, with a hardship score of 50.
3. **Ruckersville**
   Drive: about 39 minutes, with a hardship score of 50.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

WHAT COULD HELP

- Make a labor travel plan during prenatal visits, including a backup driver.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Use the WIC office in Stanardsville as a place to share birth-planning information.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 6.9 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

BIRTH CARE ACCESS · FLUVANNA COUNTY · BLUE RIDGE HEALTH DISTRICT

Fluvanna County: the south end drives the farthest

No hospital in Fluvanna County delivers babies. Martha Jefferson in Charlottesville is the closest for everyone. The drive is shorter in the north, near Palmyra and Lake Monticello, and longest around Columbia and Fork Union.

34 min
average drive to the closest hospital that delivers babies

40%
of residents live more than 30 minutes away

1 in 5
homes in Columbia/Fork Union have no internet

WHAT WE FOUND

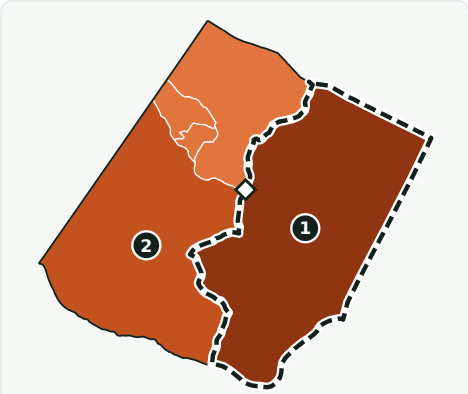
- Martha Jefferson Hospital in Charlottesville is the closest birth hospital for every neighborhood.
- The drive to WIC in Palmyra averages 18 minutes.
- The closest hospital to the south, in Farmville, stopped delivering babies in December 2025.

PLACES TO WATCH

1. **Columbia/Fork Union**
   Drive to a birth hospital: about 47 minutes. Hardship score (ADI): 54 out of 100. 22% of homes are offline. This is the county's CHIP priority area.
2. **Cunningham**
   Drive: about 45 minutes, with a hardship score of 41.

WHAT COULD HELP

- Make a labor travel plan during prenatal visits, including a backup driver.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Offer check-ins by phone for homes without internet in the south of the county.
- Hold WIC pop-up days in Fork Union or Columbia.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 7.9 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

BIRTH CARE ACCESS · ALBEMARLE COUNTY · BLUE RIDGE HEALTH DISTRICT

Albemarle County: near care, with pockets of need

Most of Albemarle is close to UVA and Martha Jefferson, the two birth hospitals in the area. Two kinds of places need attention: the far south of the county, where the drive is long, and neighborhoods near Route 29 where many families have no car.

18 min
average drive to the closest hospital that delivers babies

4%
of residents live more than 45 minutes away

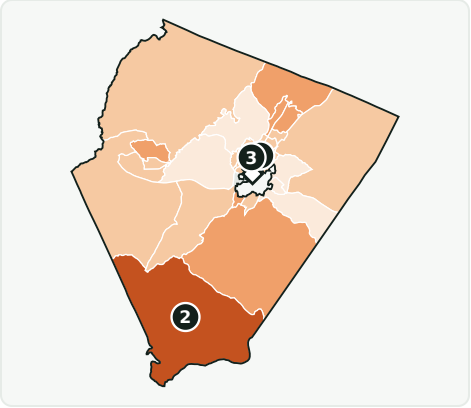
22%
of homes in Branchlands/Squire Hill have no car

WHAT WE FOUND

- UVA is the closest birth hospital for about 8 in 10 residents. Martha Jefferson is closest for most of the rest.
- WIC is at the Charlottesville/Albemarle Health Department on Rose Hill Drive. The average drive is 18 minutes.
- In Branchlands/Squire Hill, the hospital is only about 10 minutes away by car. The problem is that 1 in 5 homes has no car and 1 in 5 has no internet.

PLACES TO WATCH

1. **Branchlands/Squire Hill**
   Hardship score (ADI): 54 out of 100. Drive: about 10 minutes, but 22% of homes have no car. This is the county's CHIP priority area.
2. **Southern Albemarle-Scottsville**
   Drive: about 42 minutes, with a hardship score of 45. Not a CHIP priority area, but it has the county's highest combined score.
3. **Hydraulic**
   Hardship score of 56, with 1 in 10 homes without a car.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

WHAT COULD HELP

- In Branchlands and Hydraulic, focus on rides and bus routes to prenatal visits and WIC, not distance.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Include Scottsville in rural outreach and labor travel planning.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 6.1 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

BIRTH CARE ACCESS · CITY OF CHARLOTTESVILLE · BLUE RIDGE HEALTH DISTRICT

City of Charlottesville: close to care, but not always easy to reach

Both birth hospitals are within a few minutes' drive of every city neighborhood. Here, the barrier is not distance. It is money, cars, and internet. In Fifeville, the city's CHIP priority area, many families have no car.

5 min
average drive to the closest hospital that delivers babies

12%
of homes have no car

12%
of homes have no internet service

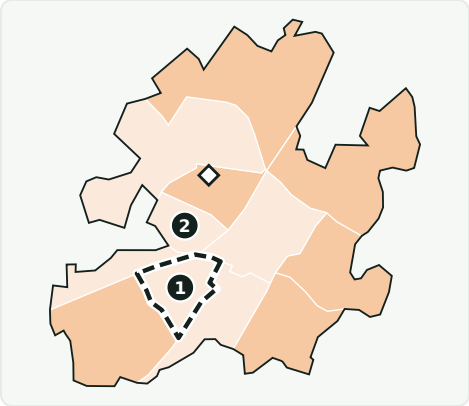
WHAT WE FOUND

- UVA is the closest birth hospital for almost 9 in 10 residents.
- WIC at the health department on Rose Hill Drive averages a 6-minute drive.
- A 5-minute drive can be a long bus trip for a family without a car, especially in late pregnancy or with a newborn.

PLACES TO WATCH

1. **Fifeville**
   Hardship score (ADI): 56 out of 100. Drive: about 2 minutes, but 16% of homes have no car and 18% have no internet. This is the city's CHIP priority area.

2. **10th & Page-Venable**
   Hardship score of 59, the highest in the city. 26% of homes have no car.



Shading: double-burden score (darker = farther from delivery care and more disadvantaged). Dashed outline: CHIP priority tract. Numbers match "Places to watch." ◆ WIC site.

WHAT COULD HELP

- Focus on rides, bus routes, and child care for visit days, not distance.
- Share how to book free Medicaid rides to prenatal and postpartum visits.
- Offer phone check-ins for families without internet.

Draft for partner review · Drafted with AI from tract data · Reading level: grade 5.8 · If labor starts fast or there is heavy bleeding, call 911.

Sources: drive times from OSRM on OpenStreetMap roads (no traffic), Sept 2026; Area Deprivation Index 2021 (Neighborhood Atlas); households without internet or a vehicle from CDC/ATSDR SVI 2022 (ACS 2018–22); population and poverty from ACS 2020–24 via UVA Equity Center; CHIP priority tracts from the BRHD 2025–2028 CHIP. "Closest hospital" is by drive time and may not be where a family delivers.

APPENDIX B

All deliverables

Everything produced for the Blue Ridge Maternal Access Map project, in one place.

Deliverable	Format and where to find it	What it contains
Interactive map	HTML link: https://claude.ai/artifact/75HKTjc2Pt58CZZTzk9wwD Private until shared from the page's Share menu.	Seven heat-map layers, sortable tract heat grid, access profiles, CHIP overlay, telehealth chart, where AI fits, one-pagers, methods and sources.
Project packet	PDF: Blue_Ridge_Maternal_Access_Project_Packet.pdf (this document)	Summary, static heat maps, profiles, CHIP overlay, telehealth, AI, methods, resume bullets, interview notes, one-pagers.
Locality one-pagers	PDF: Blue_Ridge_birth_care_one-pagers.pdf (also Appendix A here)	Six plain-language handouts (Nelson, Louisa, Greene, Fluvanna, Albemarle, Charlottesville), one letter page each, grade 6–8 reading level.
Code and data	ZIP: blue-ridge-maternal-access-map.zip	Python pipeline (data assembly, drive times, clustering, fact sheets, readability), raw and processed data, raw drive-time responses, AI drafting prompt, page source, README.
Project README	Claude project "AI-Maternal Mortality": claude/blue-ridge-maternal-access-map.md	Findings, profiles table, data sources, methods, limits, telehealth and AI sections, talking points.
Resume bullets	Section 9 of this packet	Full and short versions, telehealth and AI bullets, LinkedIn line.
Interview takeaways	Section 10 of this packet	Pitch, numbers, findings, methods in plain words, stance, limits.

Before sending the map link to anyone, open it and share it from its Share menu; until then only you can view it. Data reflect sources as of September 2026.